ANNEX
II

OBSERVATION AND CALCULATION OF LABORATORY TESTS

OBSERVATION

	Weight (kg)										
	Sample + Tray	Sample + Mould	Sample before oven dry	Sample after oven dry	Mould	Tray	Sample for water added	Sample for density	Moisture content (%)	Density, g/cm^3	Dry density (yd), g/cc
Sample 1	10.46	10.9	0.084	0.072	6.7	4.12	6.34	4.2	14.28571429	1.827539055	1.599097
Sample 2	10.2	10.5	0.058	0.05	6.7	4.12	6.08	3.8	13.79310345	1.653487717	1.453065
Sample 3	9.62	11.18	0.174	0.146	6.7	4.12	5.5	4.48	16.09195402	1.949374993	1.679165
Sample 4	9.92	9.35	0.154	0.13	5.3	4.12	5.8	4.05	15.58441558	3.268895236	2.828145
Sample 5	10.12	9.32	0.194	0.162	5.3	4.12	6	4.02	16.49484536	3.244681197	2.785257
Sample 6	10.32	10.6	0.094	0.08	6.7	4.12	6.2	3.9	14.89361702	1.697000552	1.477019
Sample 7	12.8	10.42	0.032	0.028	6.7	4.12	8.68	3.72	12.5	1.618677449	1.438824

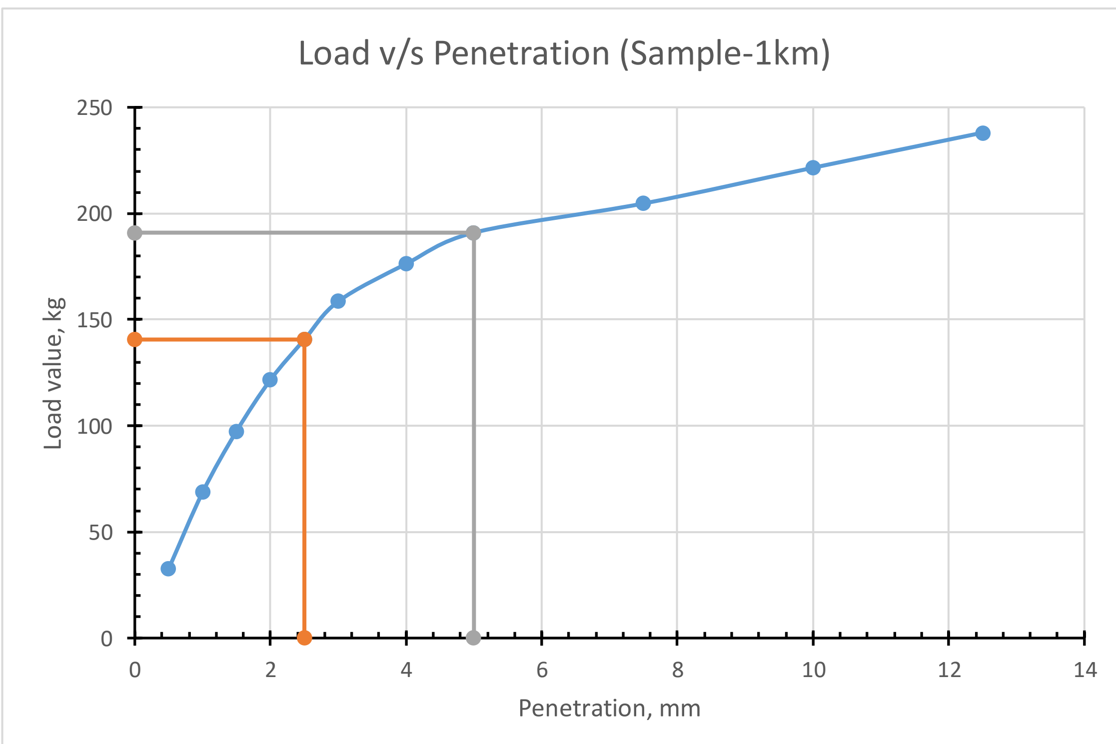
Volume of 5.3kg mould=	1238.950688	cm^3
Volume of 6.7kg mould=	2298.1725	cm^3

Observation:	Value
Sample	Sample 1
Weight of hammer	4.86
Blows per layer	56
No of layers	3
Weight of the mould	6.7kg
Volume of mould	2298.1725 cm^3
Sample Calculation	
Test no.1	
Weight of soil + mould:	10.9 kg
wt. of soil	4.2 kg
Density of soil(γ)	1.827539 g/cm^3
For water content	
Wt. of wet soil(w1)	84 gm
wt. dry soil(w2)	72 gm
wt. of water(w3)	12 gm
Water content(w)	(w3/w2) × 100 = 14.286 %
Dry density, (γ d)	γ / (1+w) = 1.6 gm/cc

CBR TESTING OF SAMPLE

Sample 1 (1+170 km)

PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	20	100	32.7
1	42	210	68.67
1.5	59.6	298	97.446
2	74.4	372	121.644
2.5	86	430	140.61
3	97	485	158.595
4	107.8	539	176.253
5	116.8	584	190.968
7.5	125.2	626	204.702
10	135.5	677.5	221.5425
12.5	145.6	728	238.056

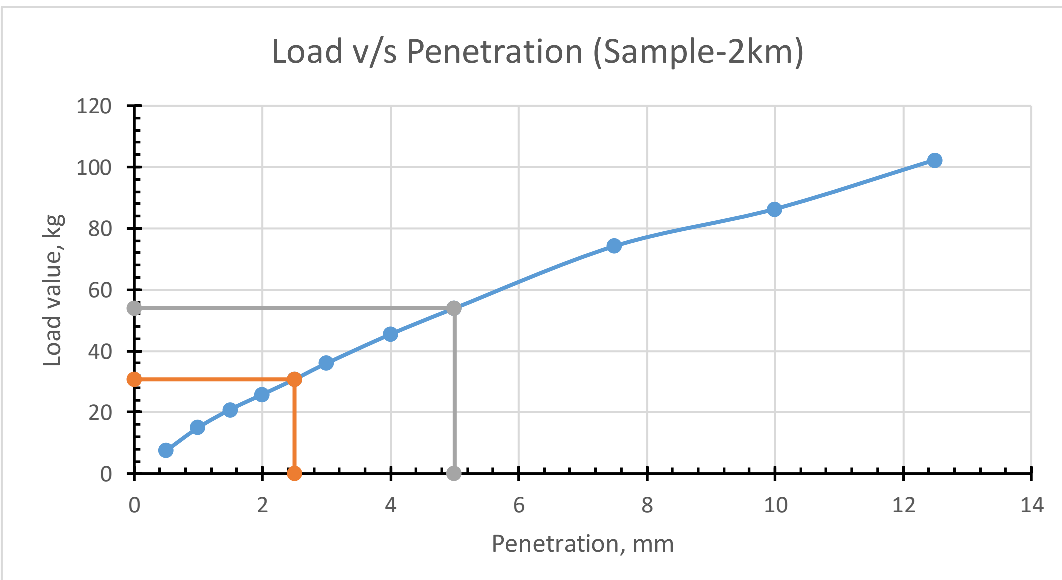


From Graph, Load corresponding to 2.5 mm penetration =140.61kg.
Therefore, CBR 2.5 mm = (140.61 ×100) / 1370 = 10.26 %
Load corresponding to 5 mm penetration =190.968 kg.
Therefore, CBR 5 mm = (190.968×100) / 2055 =9.29%.

As CBR 2.5 mm > CBR 5 mm, hence CBR value of the given soil sample = 10.26 %

<u>Sample 2</u> (2+480 km**)**

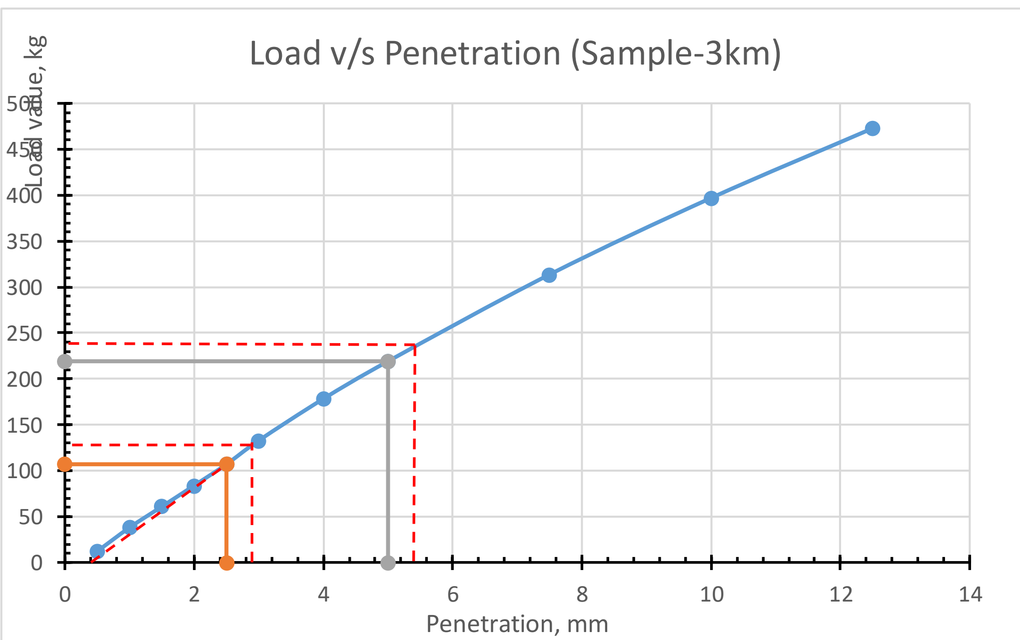
PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	4.6	23	7.521
1	9.2	46	15.042
1.5	12.8	64	20.928
2	15.8	79	25.833
2.5	18.8	94	30.738
3	22	110	35.97
4	27.8	139	45.453
5	33	165	53.955
7.5	45.4	227	74.229
10	52.8	264	86.328
12.5	62.6	313	102.351



At 2.5 mm, CBR% = 2.24 %
At 5 mm, CBR% = 2.625%
CBR of sample 2= 2.625 %

<u>Sample 3</u> (3+100 km)

PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	7.4	37	12.099
1	23.2	116	37.932
1.5	37.2	186	60.822
2	51.2	256	83.712
2.5	65.4	327	106.929
3	81.2	406	132.762
4	109	545	178.215
5	134	670	219.09
7.5	191.8	959	313.593
10	242.8	1214	396.978
12.5	289.2	1446	472.842

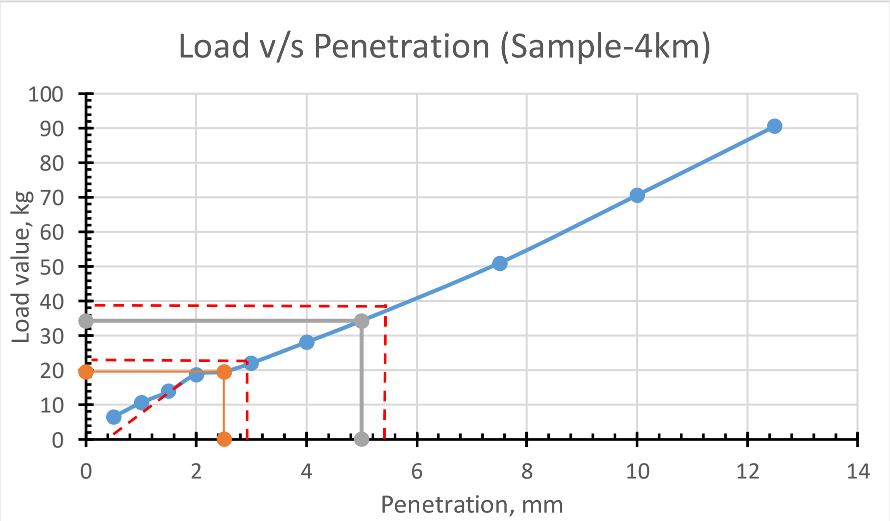


Sample 3 CBR correction:

δ	0.4mm		
corrected penetration		2.5+0.4 =	2.9mm
		5.0+0.4 =	5.4mm
corrected load from graph for			
		2.9mm	127.595 kg
		5.4mm	234.21 kg
corrected CBR at			
	2.9 mm	9.313%	
	5.4 mm	11.397%	

Sample 4 (3+960 km)

PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	4	20	6.54
1	6.5	32.5	10.6275
1.5	8.5	42.5	13.8975
2	11.5	57.5	18.8025
2.5	12	60	19.62
3	13.5	67.5	22.0725
4	17.2	86	28.122
5	21	105	34.335
7.5	31.2	156	51.012
10	43.2	216	70.632
12.5	55.4	277	90.579

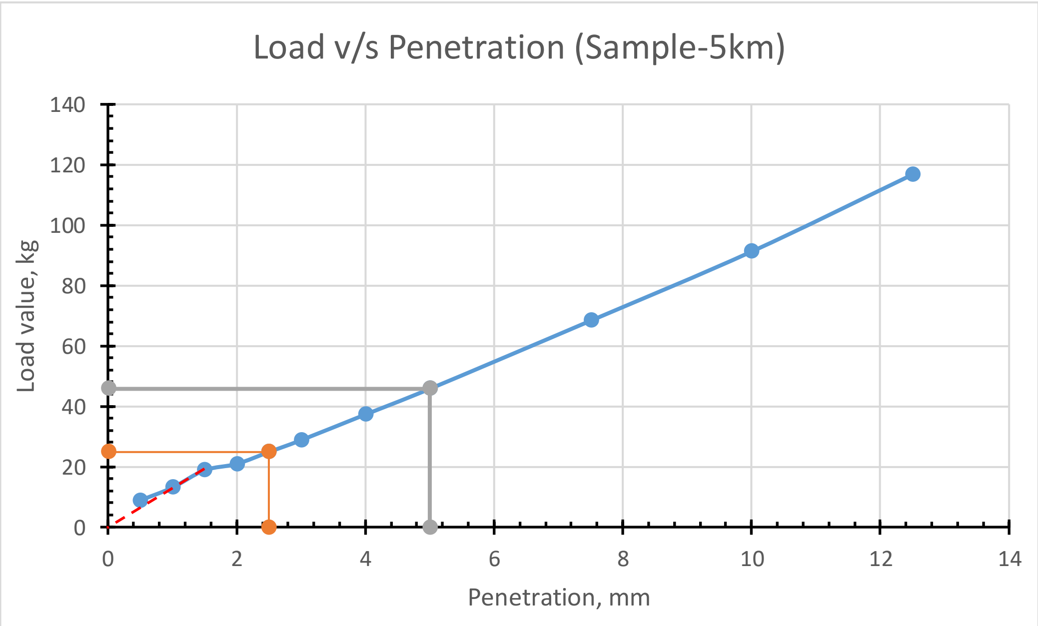


Sample 4 CBR correction :

δ	0.3mm			
corrected penetration		2.5+0.3 =	2.3mm	
		5.0+0.3 =	5.3mm	
corrected load from	graph for			
		2.2 mm	21.582	kg
		4.7mm	33.336	kg
corrected CBR at				
	2.8 mm	1.575%		
	5.3 mm	1.622%		

Sample 5 **(4+790 km)**

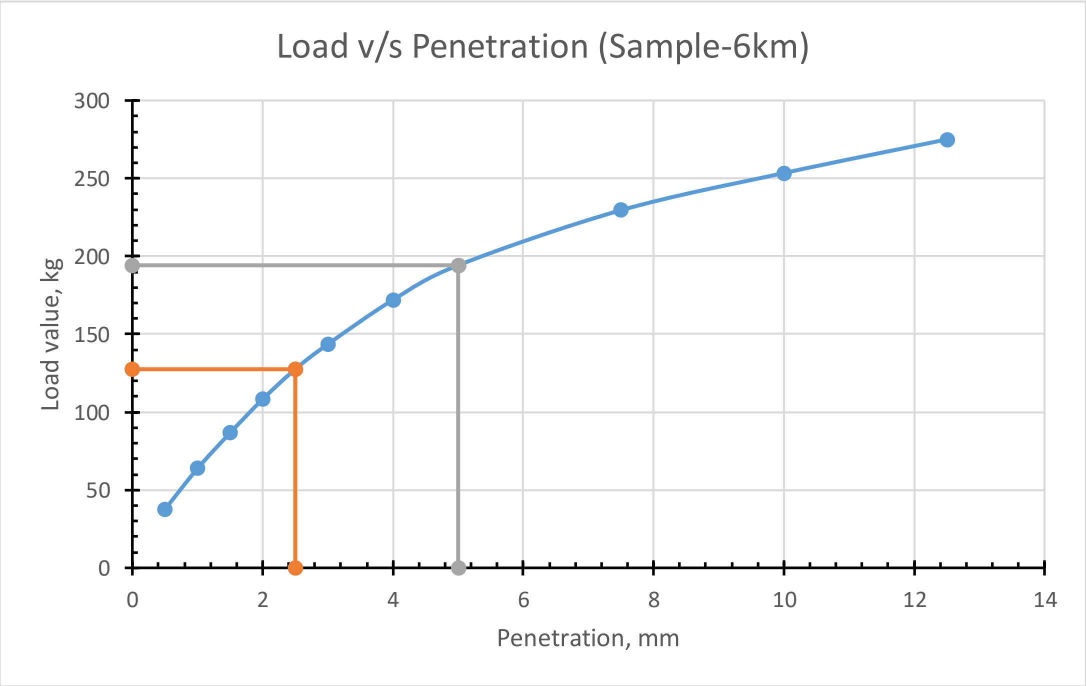
PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	5.4	27	8.829
1	8	40	13.08
1.5	11.5	57.5	18.8025
2	12.8	64	20.928
2.5	15.2	76	24.852
3	17.6	88	28.776
4	22.8	114	37.278
5	28	140	45.78
7.5	41.8	209	68.343
10	55.8	279	91.233
12.5	71.4	357	116.739



$CBR_{2.5}$=	1.814 %
CBR_5=	2.23 %

Sample 6 (5+580 km)

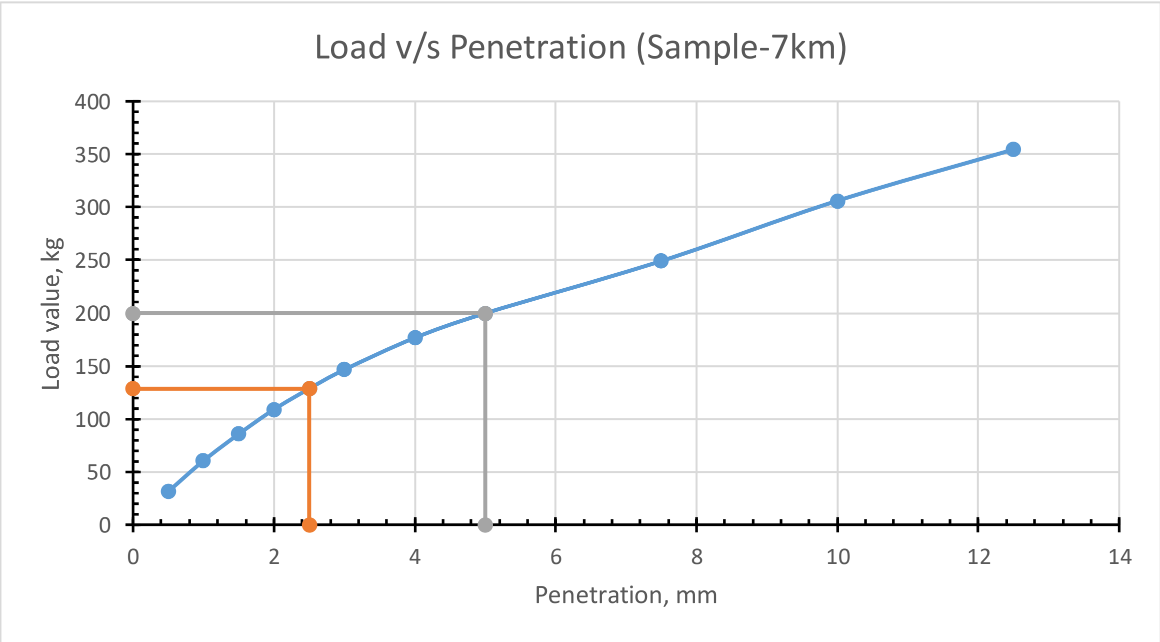
PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	23	115	37.605
1	39	195	63.765
1.5	53	265	86.655
2	66.2	331	108.237
2.5	78	390	127.53
3	87.8	439	143.553
4	105.2	526	172.002
5	118.8	594	194.238
7.5	140.4	702	229.554
10	155	775	253.425
12.5	168.2	841	275.007



$CBR_{2.5}$=	9.308 %
CBR_5=	9.452 %

<u>Sample 7</u> (6+820 km)

PENETRATION (MM)	PROVING RING READING (MM)	DIVISION	LOAD(KG)=DIVISION*PROVING RING CONSTANT
0.5	19.2	96	31.392
1	37	185	60.495
1.5	52.4	262	85.674
2	66.6	333	108.891
2.5	78.6	393	128.511
3	89.6	448	146.496
4	108	540	176.58
5	122.2	611	199.797
7.5	152.4	762	249.174
10	187.2	936	306.072
12.5	216.8	1084	354.468



$CBR_{2.5}$=	9.38 %
CBR_5=	9.72 %

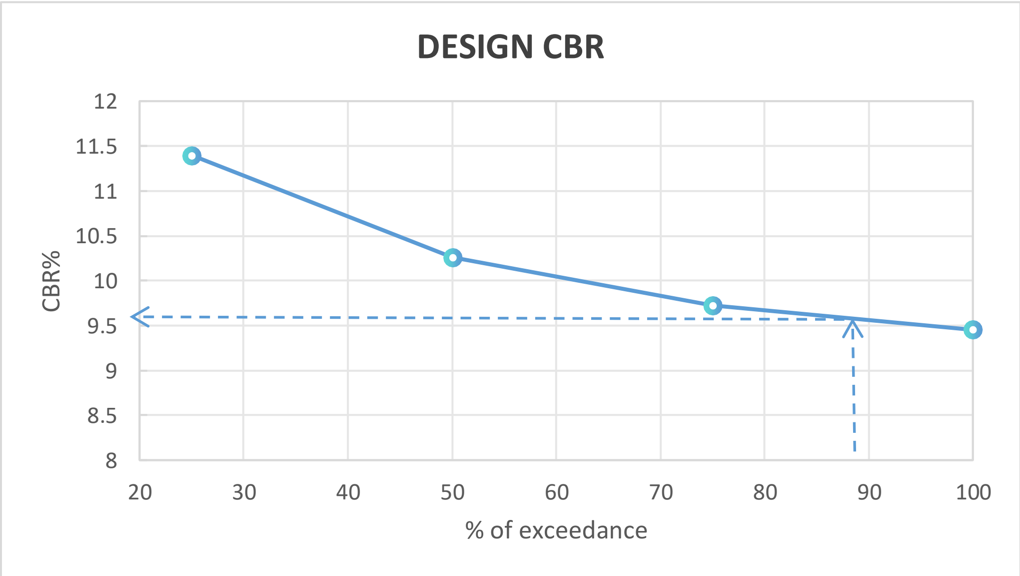
Classiifying CBR results,

CBR Value below 5%	CBR value above 5%	% of exceedance	Rank
2.625	11.397	25	1
2.227	10.263	50	2
1.622	9.722	75	3
	9.452	100	4

By Asphalt Institute Method:

ESALs	Percentile to select the CBR design value
$\leq 10^4$	60.0
$10^4 - 10^6$	75.0
$\geq 10^6$	87.5

ESAL value is 116601.818 standard axle.Based on the traffic level, a 75% exceedance would normally be adequate as per Asphalt Institute method. However, we adopted 87.5% to reflect variability in subgrade strength along the alignment and the limited number of test samples. This yields a slightly lower design CBR and a modestly thicker pavement. Traffic loading is already accounted for through ESAL, so this choice specifically improves reliability of the subgrade input. Overall, it is a deliberate, conservative selection to ensure consistent field performance.



Thus, Percentile to select the CBR design value is 87.5th percentile. We get, CBR% (>5%) in 87.5th percentile equals to 9.58%.And CBR(<5%) equals to1.622% by assuming lowest CBR for higher pavement thickness.